

Short communication

Novel battery management systems: Enhancing flexibility and fault tolerance

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A B S T R A C T

The battery management system (BMS) is crucial for the functionality of battery-powered systems, providing flexibility and safety under nominal operating conditions. However, when operating beyond nominal conditions, BMS becomes susceptible to faults, leading to potential system failures. To address this issue, this paper proposes the design and development of a flexible and fault tolerant modular BMS employing a hierarchical isolation approach. The proposed BMS comprises two main modules: the central management unit and the local management unit. It incorporates three levels of isolation: module-level, component-level, and ground-level isolation, ensuring robust performance even in faulty conditions. Simulation and actual prototype of the proposed BMS are presented and discussed, including testing with various faulty scenarios. Results indicate a mean absolute error of 0.0015 V between measured and actual voltage monitoring values, demonstrating a significant improvement of 98.92 % in performance compared to conventional BMS under faulty conditions.

Keywords

Battery management system

Fault tolerance

Hierarchical modularity

Lithium-ion

Modular

1. Introduction

The increasing reliance on battery-based applications has encouraged ongoing innovations in battery management systems (BMS) [1,2]. Traditional BMS advancements have primarily concentrated on performance under normal operating conditions, often overlooking the situations of flexibility and safety during failures. [3] presents a scalable and decentralized framework aiming at flexible deployment, yet it neglects the critical aspect of safety. While [4] elaborates on BMS design encompassing cell balancing, monitoring, and circuit protection, it omits adherence to established safety design standards. Furthermore, the open-source BMS discussed in [5] shows compatibility with varied battery cell types but fails to address scalability, adaptability, and fault tolerance. Common causes of BMS failure include 1) cell imbalance, 2) sensor malfunctions, 3) communication errors, 4) component degradation, 5) electromagnetic interference, 6) overheating, 7) isolation breakdown, and 8) power supply issues. Any of these faults may lead to a module malfunction, ultimately causing the module to stop functioning.

The work presented in [6] accommodates hardware and software frameworks for an adaptable BMS under normal circumstances, disregarding the essential considerations for faulty scenarios. On the other hand, the modular approach in [7], tailored for energy storage systems (ESS), concentrates mostly on reducing operating costs, sidelining the crucial element of fault tolerance which significantly impacts design-cost dynamics. Moreover, [8,9] explores BMS functionalities under severe conditions like high voltage and overcurrent faults within electric vehicle applications, yet they lack comprehensive discussions on safety and adaptability aspects of BMS design.

Despite significant improvements toward enhancing BMS configurations [10–13], there remains an obvious gap in addressing flexibility and safety during faulty conditions. Existing works are aiming on maintaining the stability of the BMS such as using fault localization. Fault localization is carried out by estimating the state of charge (SOC) of individual cells and utilizing a cell balancing strategy. Cells that fail to maintain the charge cycle or discharge beyond their depth of discharge are monitored and flagged after a predetermined time-period. This process is crucial for onboard diagnostics, as identifying faulty cells helps reduce diagnosis time, minimize the risk of thermal runaway, and lower maintenance efforts, ultimately improving system reliability [14]. [15] investigates how integrating redundant submodules into modular multilevel converter (MMC)-based battery energy storage systems (BESSs) can enhance fault tolerance. Redundant Submodules are extra submodules added to each arm of the MMC to ensure continued

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Received 15 April 2024; Received in revised form 30 October 2024; Accepted 30 November 2024

Available online 14 December 2024

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operation even if some submodules fail. They help maintain the system's performance by providing additional flexibility and fault tolerance. However, as the rate of improvement decreases with the addition of more redundant submodules, implementing many redundant SMs may not be justified in achieving the desired level of battery fault tolerance.

In [2], an isolated active balancing and monitoring system for Lithium-Ion Battery stacks utilizing a single transformer per cell. The isolation method allows for accurate cell voltage monitoring and active cell balancing. It ensures safety by preventing any direct electrical connection between the high voltage battery pack and the monitoring system. Using a silicon diode for isolation causes a voltage drop, lowering the efficiency of energy transfer. While a Schottky diode could enhance efficiency due to its lower forward voltage drop, it would lead to increased reverse leakage current, particularly at elevated temperatures.

Though and all, limitations are still present to existing works focusing on BMS fault isolation. The proposed hierarchical isolation presented in this paper introduces a novel design approach for modular BMS considering fault tolerance, extending beyond our initial proposal in [1]. The fault considered in this paper specifically focuses on the scenario where a local management unit (LMU) module completely stops functioning. This is treated as the worst-case fault scenario and can be caused by several contributing factors, including sensor malfunctions, communication errors, component degradation, electromagnetic interference, overheating, or isolation breakdown. While unilateral LMU failure is a critical aspect, we acknowledge that other fault conditions could similarly lead to BMS failure. This innovative BMS architecture integrates dual module type which realizes a modular architecture: the central management unit (CMU) and LMU, to improve flexibility. Additionally, it adopts a tiered/hierarchical approach to safety, featuring three levels of isolation: 1) module-level, 2) component-level, and 3) ground-level, thereby ensuring comprehensive protection.

The contributions of this paper include the following:

- **Novel Hierarchical Isolation Structure:** The proposed design introduces a hierarchical isolation structure, separating the BMS into three distinct levels: 1) module, 2) component, 3) and ground. This multi-tiered approach optimizes system flexibility and safety, particularly under faulty conditions, by isolating faults within individual modules and preventing their propagation throughout the system.
- **Comparison with Conventional Systems:** Unlike conventional systems that typically implement isolation at a single layer, the proposed hierarchical approach significantly enhances fault resilience. By enabling targeted isolation at multiple points (module, component, and ground levels), the system improves fault containment and reduces electrical interference, leading to increased reliability and robustness, especially in high-power or safety-critical applications.
- **Validation through Simulation and Empirical Evaluation:** The effectiveness of the hierarchical isolation design has been validated through simulations and empirical evaluations. Voltage measurement accuracy is used as a key performance metric to demonstrate the efficacy of signal and power transfer isolation. The results, presented in [Tables I and II](#), and supported by the waveform in [Fig. 8](#), confirm that the proposed BMS design offers superior performance under fault conditions, ensuring stable operation and improved fault tolerance.

Concisely, the proposed BMS structure advances the field by providing enhanced safety, reliability, and fault tolerance through its multi-tiered isolation approach, where module-level, component-level, and ground-level isolation mitigate fault propagation, reduce electrical interference, and ensure stable operation across different fault conditions, making it ideal for high-power or safety-critical applications.

Table I

Battery voltage cross-performance fault analysis at 14.4 (V).

1 Faulty LMU				
Fault	LMU 1			Average voltage difference: 0.002 V
Active LMU module	LMU 2	LMU 3	LMU 4	
Actual module voltage (V)	14.379	14.352	14.378	
Monitored module voltage (V)	14.377	14.352	14.377	
Voltage difference	0.002	0.00029	0.0012	

2 Faulty LMUs				
Fault	LMU 1 and LMU 2			
Active LMU module	LMU 3	LMU 4		
Actual module voltage (V)	14.412	14.349		Average voltage difference: 0.001 V
Monitored module voltage (V)	14.412	14.349		
Voltage difference	0.00018	0.00012		

3 Faulty LMUs				
Fault	LMU 1, LMU 2 and LMU 3			Average voltage difference: 0.002 V
Active LMU module	LMU 4			
Actual module voltage (V)	14.381			
Monitored module voltage (V)	14.381			
Voltage difference	0.00011			

Table II

Comparative advantage in terms of voltage difference (V).

Reference	Average voltage monitoring difference at no fault condition (V)
[10]	0.058
[11]	0.541
[12]	0.131
[13]	0.01
[Proposed]	Average voltage monitoring difference when subjected to faulty condition (V) 0.002

2. Proposed BMS hierarchical modularity

The proposed modular BMS enhances flexibility by allowing multiple LMUs to connect to a single CMU, improving system adaptability. Safety is strengthened through a hierarchical isolation design applied to the modular BMS, which not only improves fault isolation but also enhances grounding integrity. The proposed hierarchical isolation technique includes three levels: 1) module-level, 2) component-level, and 3) ground (GND) level isolation. These layers inherently reduce interference during faults and regular operation.

- **Module-level isolation:** This separates individual battery modules (LMUs), effectively blocking noise pathways and minimizing interference between them.
- **Component-level isolation:** This shields sensitive components like sensors and controllers from electrical noise, ensuring that signal quality remains unaffected by external interference.
- **GND-level isolation:** This decouples the BMS ground from the common ground noise, breaking the loop through which common-mode noise could propagate. By isolating the ground, the system prevents common-mode signals from causing interference in the system's operation.

These mechanisms ensure the BMS remains stable and resistant to interference, even in noisy operating environments. In contrast,

traditional single-layer isolation systems typically require a full shut-down of the entire BMS during fault conditions, resulting in elevated repair and operational costs. By implementing isolation at multiple levels, the proposed system enhances fault tolerance and mitigates cross-interference between modules, leading to greater overall system reliability and robustness. Additionally, the improved fault containment and safety features reduce the risk of large-scale failures, which are considerably more costly to address. As a result, despite a marginal increase in upfront manufacturing costs, the hierarchical isolation design proves to be substantially more cost-effective over the operational lifetime of the BMS due to its ability to minimize both repair expenses and operational disruptions. In the operating case scenario presented in the results and discussion, the fault simulated is a critical failure wherein the LMU module completely stops functioning. This failure was used in simulation, testing, and certification to validate the robustness of the proposed hierarchical isolation design. The proposed modular topology facilitates the isolation of the faulty module of both CMU and LMU via its hierarchical isolation scheme. This would allow the remaining modules such as the LMUs to maintain functionality while the issue is diagnosed and resolved. Our hierarchical isolation structure further enhances this strategy by containing faults at multiple levels, module, component, and ground; thereby ensuring greater system reliability and minimizing downtime, positioning it as an advanced solution for effective fault management.

The proposed isolation technique can be quantitatively represented by the following equations, providing a clear analysis of system flexibility and fault tolerance. The system flexibility (F_{SYS}) is influenced by the number of LMUs (N_{LMU}) connected to the CMU and the degree of isolation (S_{ISO}). This relationship is defined by Eq. (1):

$$F_{SYS} = N_{LMU} \times S_{ISO} \quad (1)$$

For this paper, we use one CMU and four LMUs to demonstrate the effectiveness of the proposed system. The isolation factor safety (S_{ISO}), which represents the degree of fault isolation and interference reduction, is the sum of the isolation at the module (I_{MOD}), component (I_{COMP}), and ground (I_{GND}) levels, as shown in Eq. (2):

$$S_{ISO} = I_{MOD} + I_{COMP} + I_{GND} \quad (2)$$

These equations provide a structured way to evaluate how different levels of isolation contribute to the overall safety and performance of the BMS. By quantifying the isolation at multiple levels, the system achieves enhanced fault tolerance and reduced interference, ensuring that the BMS operates reliably under varying conditions. This mathematical framework supports the effectiveness of the hierarchical isolation design, reinforcing its adaptability and robustness in practical applications.

The fault simulated is a critical fault wherein the LMU module had fully stopped working. This fault is used for the simulation, testing and actual certification of the prototype used to prove the effectiveness of this hierarchical isolation approach. The prototype is presented in our

previous works with certification achieved. The hierarchical structure consists of three levels of isolation to prevent the spread of faults, detailed in Fig. 1(a). This design aligns with ISO26262 safety standards, mirroring the approach used in [2]. The highest level of isolation separates the CMU and LMUs, ensuring that a problem in one does not affect the other modules and its subcircuits. Within these units, there is further isolation among various components responsible for functions such as protection, monitoring, signal processing, voltage translation, digital isolation, and balancing. Thus, if a subcircuit experiences an issue like overcurrent or overvoltage, the rest of the system remains unaffected, owing to digital and ground isolators between modules. This setup not only preserves signal and power integrity under fault conditions but also lowers maintenance costs. The modular isolation, illustrated in Fig. 1(b) in detail, ensures that faults within one LMU do not impact others, promoting an uninterrupted system operation.

A. Central management unit (CMU)

The CMU module is illustrated in Fig. 2. The CMU acquires and processes the data from the connected LMUs and consists of four main components: 1) current measurement IC (INA253-Q1), 2) charge-discharge IC (BQ76200), 3) Wi-Fi module (CCS3220SF), and 4) central MCU (F28379D C2000). The central MCU is the main component where control is deployed which has GND level and component level isolation when communicating with the LMUs. Specifically, Fig. 2(a) presents the 3D layout of the CMU, showcasing its primary components which include communication, isolation, signal processing, control systems, and charge-discharge circuits. Fig. 2(b) depicts the X-ray image of the level shifter, which plays a crucial role in facilitating communication and ensuring protective isolation between the CMU and LMUs. Utilizing the level shifter further allows for the isolation between the grounds of the CMU and LMU, thereby enhancing the safety and reliability of the proposed BMS. Fig. 2(c) illustrates a block diagram representing how signals are translated between the CMU and LMU through the level shifter. It indicates that the operating voltage for the CMU is 3.3 V, whereas the LMU operates at 5 V. Despite these differences in operating voltages, effective communication is maintained through the level shifter, which adjusts signals from 3.3 V to 5 V and vice versa. In the design presented in this paper, the CMU is configured to support four LMUs. Consequently, it incorporates four level shifters to manage independent communication and isolation for each LMU.

B. Local management unit (LMU)

The LMU module is depicted in Fig. 3. Fig. 3(a) presents the 3D layout of the LMU, showcasing its principal components, including those for communication, isolation, signal processing, control, and balancing. In terms of signal processing, which is directly influenced by data acquisition frequency and resolution, the primary focus is on monitoring the battery pack's voltage and current. The battery monitoring is

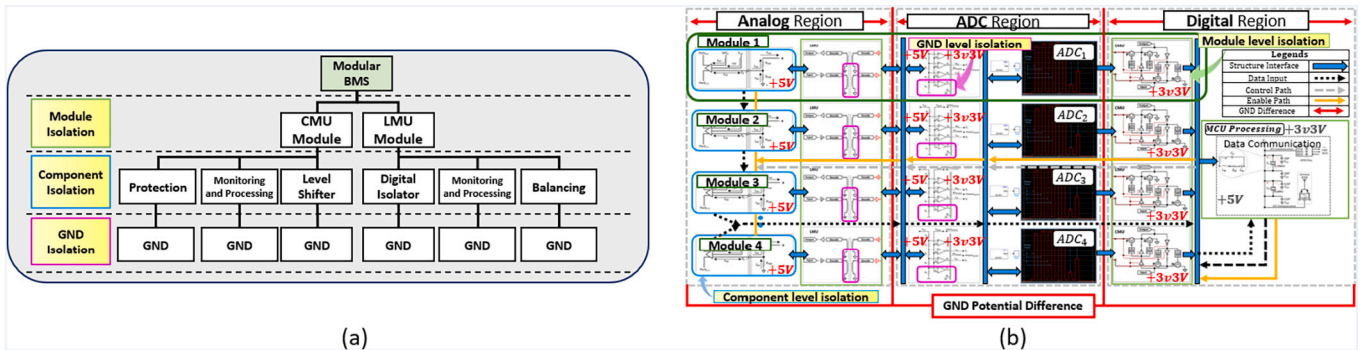


Fig. 1. The proposed (a) BMS hierarchical model (b) detailed overview of the proposed modular BMS.

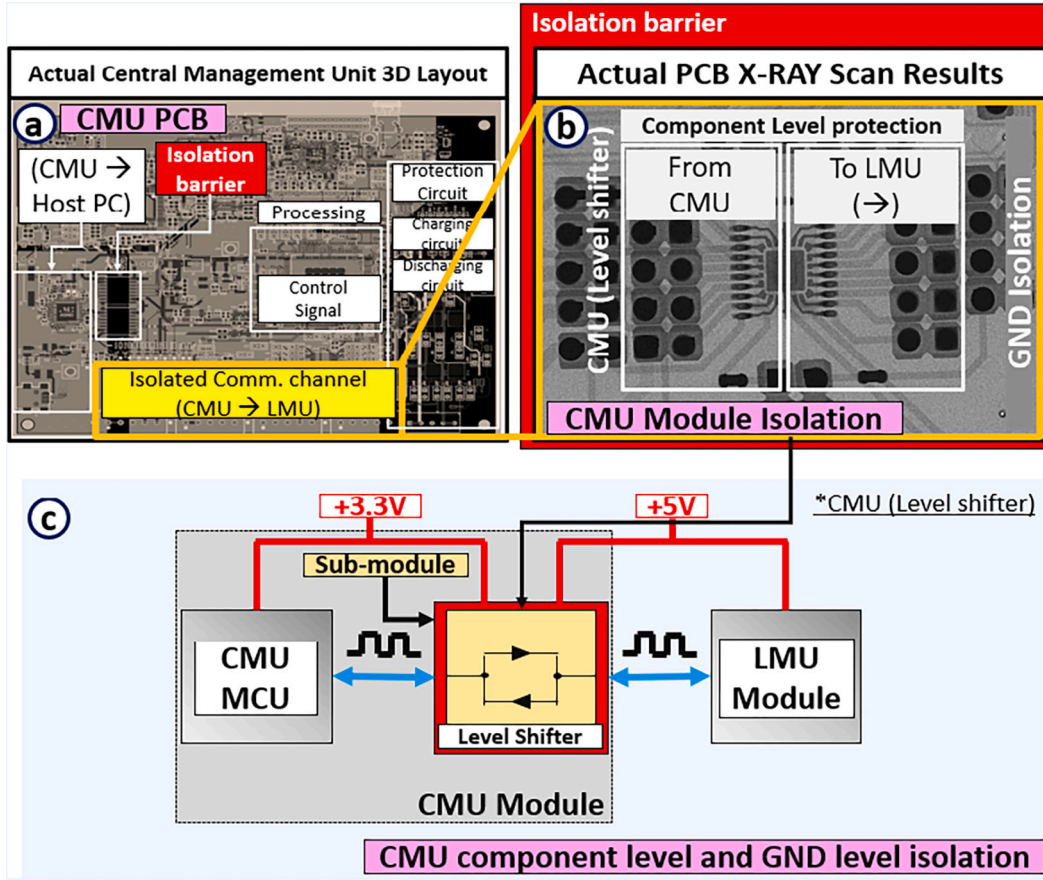


Fig. 2. Proposed CMU (a) CMU 3D layout, sub-circuits of the CMU module (b) CMU level shifter x-ray results and (c) analysis of CMU communication (level shifter).

managed by the CMU, responsible for two key tasks: 1) battery pack voltage monitoring and 2) current measurement.

- **Voltage Measurement:** The overall battery pack voltage is determined by summing the voltages from individual modules monitored by the LMUs. Each module voltage is sampled by a 12-bit Analog-to-Digital Converter (ADC), which provides a high-resolution measurement of 1.5 mV per LSB. LSB is the least significant bit voltage. This precision ensures accurate monitoring of voltage signals, even in fault-tolerant scenarios.
- **Current Measurement:** For current measurement, the CMU integrates a current measurement IC, which converts the battery current into an equivalent voltage signal. This signal is processed by the CMU's 12-bit ADC, offering precise current acquisition. The conversion from the digital reading to the corresponding analog voltage is handled using Eq. (3):

$$V_a = \frac{V_{ref}}{2^n - 1} V_d \quad (3)$$

where V_a represents the equivalent analog voltage, V_d is the corresponding digital voltage, V_{ref} is the reference (set to 2.5 V), and n denotes the ADC resolution (12 bits). For example, a digital value of 4259 converts to an analog voltage of 2.6 V. The equivalent analog voltage is then converted to the corresponding current value using Eq. (4):

$$I(A) = \frac{V_a - 2.5}{100mV} \quad (4)$$

where $I(A)$ is the pack current and V_a is the equivalent analog voltage. This method ensures high-resolution current measurement, maintaining fault tolerance in challenging conditions. The equivalent analog voltage in the given example is converted to pack current value using Eq. (4) as

1 A (A).

- **Temperature Measurement:** Although not the primary focus in this paper, the temperature measurement system operates with similar ADC precision, ensuring accurate readings under varying thermal conditions to maintain system integrity in fault-tolerant operations. These processes ensure the BMS provides high-resolution, accurate data acquisition, even in fault-tolerant scenarios, supporting reliable system performance.

The presented modular BMS integrates a comprehensive battery monitoring IC that utilizes a sampling system for ADC conversion. It achieves an average conversion time of 0.5 ms, provided all signals are clearly distinguishable, in line with the sigma-delta modulator's 512 kHz rate. A built-in voltage reference with 8 ppm/°C temperature coefficient, combined with the ADC, provides the LTC6803 with exceptional measurement accuracy. The sigma-delta ADC outputs a 12-bit code with an offset value of 0×200 (512 in decimal). D_{OUT} is the digital output represented as a decimal integer, while V_{LSB} refers to the least significant bit voltage, valued at 1.5 mV. The RC filtering circuit consists of a series resistor and a shunt capacitor, providing 30 dB of attenuation whereas these values are designed for 67.2 V battery pack load utilization. Fig. 3 (b) displays the X-ray image of the dual-channel digital isolator, which plays a crucial role in providing protective isolation for the LMU from the CMU. This isolation allows the LMU to operate independently from other LMUs, thereby facilitating a modular configuration. Fig. 3(c) illustrates a block diagram highlighting the ground isolation between the CMU and LMU, which is critical as the modular configuration results in LMUs operating at different voltage levels. Despite these voltage discrepancies, as depicted, the LMU can maintain safe communication with the CMU. Furthermore, should any LMU encounter a failure, the digital

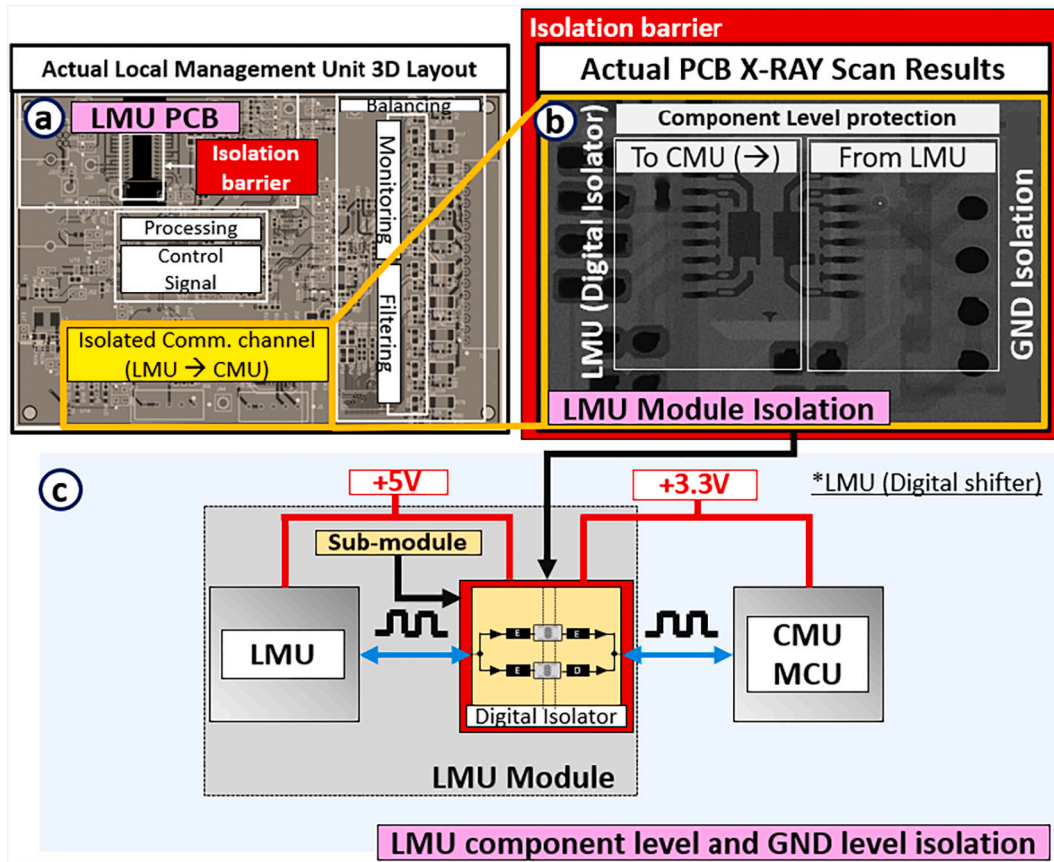


Fig. 3. Proposed LMU (a) LMU 3D layout, sub-circuits of the LMU module (b) LMU level shifter x-ray results and (c) analysis of LMU communication (digital isolator).

isolator intercedes to isolate the fault, ensuring it does not impact the CMU or other LMU modules.

Fig. 4 illustrates the isolation strategies employed within the BMS. The module-level isolation separates the LMUs from the CMU, representing the initial layer of BMS fault protection. The component-level isolation distinguishes between components of the CMU and LMU. Finally, ground-level isolation forms the foundational layer of our

hierarchical isolation strategy in the proposed BMS architecture.

In the implementation and performance evaluation of the proposed BMS, four LMUs are connected to one CMU. To evaluate whether the proposed configuration maintains performance under various fault conditions, we introduced different fault scenarios into the system. Additionally, we evaluated the flexibility and safety under various operational conditions, including different total battery pack voltages: minimum, nominal, and maximum. For performance evaluation, we utilized a battery pack with a nominal voltage of 67.2 V, divided into four LMUs. Each LMU is designed to support up to 12 cells and operates within a voltage range: 12.8 V (minimum), 14.4 V (nominal), and 16.8 V (maximum). Additionally, the individual cells within an LMU are defined to have voltages of 3.2 V (minimum), 3.6 V (nominal), and 4.2 V (maximum).

3. Results and discussion

In the implementation and performance evaluation of the proposed BMS, four LMUs are connected to one CMU, and the system is tested with up to four faulty LMUs as a test scenario. The fault design principle defines critical failure as the scenario in which the LMU module ceases all functionality due to a fault, as shown in Fig. 5. This complete halt is a severe condition, triggered when the LMU is subjected to faults that disrupt its ability to operate, rendering it inactive. The design approach ensures that the system prioritizes isolating critical failures. This fault scenario gives emphasis on the importance of stability and fault tolerance in situations where the LMU encounters operational interruptions or potential damage. Fig. 5 depicts the various faulty scenarios utilized to evaluate the system performance. Table I summarizes the outcomes. The findings show minimal variance in voltage measurement accuracy, regardless of the presence of one, two, or three faulty LMUs. This

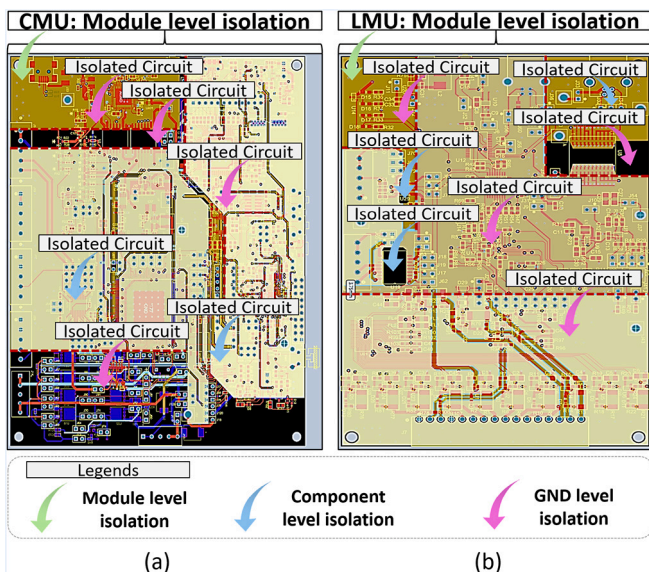


Fig. 4. Isolation of each BMS sub-circuit (a) CMU (b) LMU.

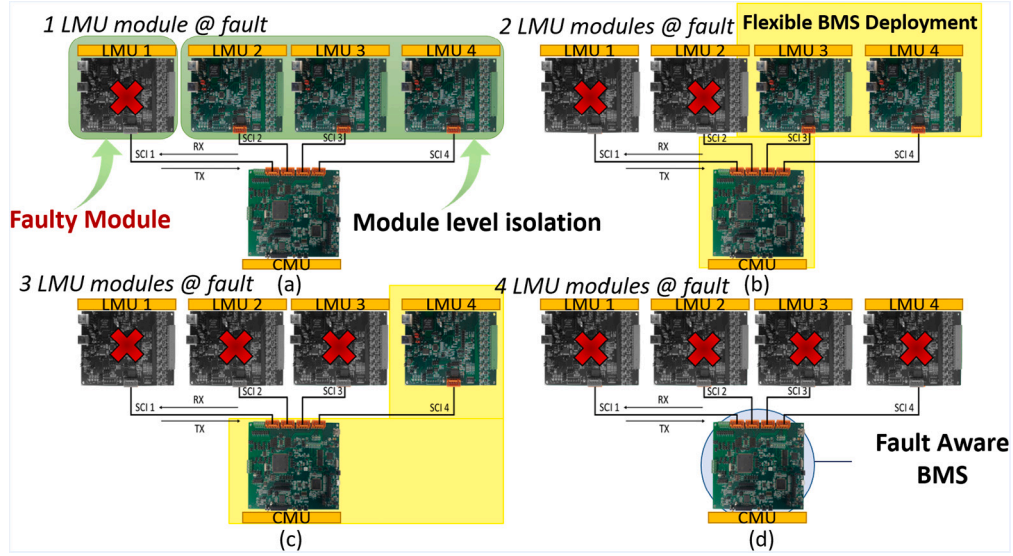


Fig. 5. The possible fault scenarios in BMS (a) 1 LMU at fault (b) 2 LMUs at fault (c) 3 LMUs at fault and (d) 4 LMUs at fault.

suggests that module faults do not significantly impact the performance of the remaining LMUs or the overall system. In Fig. 6, the proposed BMS is compared with other benchmark BMS designs, referenced in [10–13], focusing on the average voltage difference between actual and monitored values. Given that this paper introduces a novel BMS with a hierarchical isolation design, we have expanded the comparison to include additional references to recent BMS designs, including the widely adopted master-slave architecture. Our comparison focuses on key performance metrics such as voltage difference in normal operating conditions and fault tolerance. These results demonstrate the superior fault tolerance and stability of the proposed BMS compared to existing systems. [10–13] especially [13] gave emphasis that the critical parts of a BMS are the input voltages and currents measurement circuits hence this gives justification on the main performance metrics used in this paper which is the voltage difference.

Fig. 7 shows the actual experimental setup used to evaluate the implementation viability and fault tolerance of the proposed modular BMS under various conditions. Fig. 7(a) depicts the data acquisition and verification process, while Fig. 7(b) illustrates the modular BMS connection, highlighting the CMU and LMU modules. The experimental tests were conducted across a range of temperatures, from -20°C to 60°C , to assess the BMS's fault tolerance under extreme conditions. Load

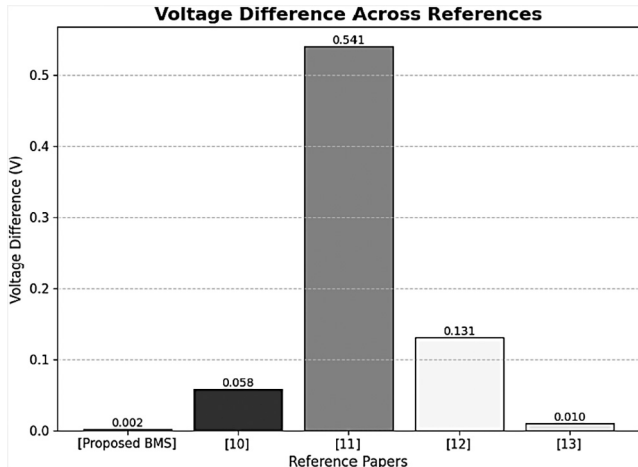


Fig. 6. The voltage monitoring accuracy of the proposed modular BMS versus the voltage monitoring accuracy of reference [10–13].

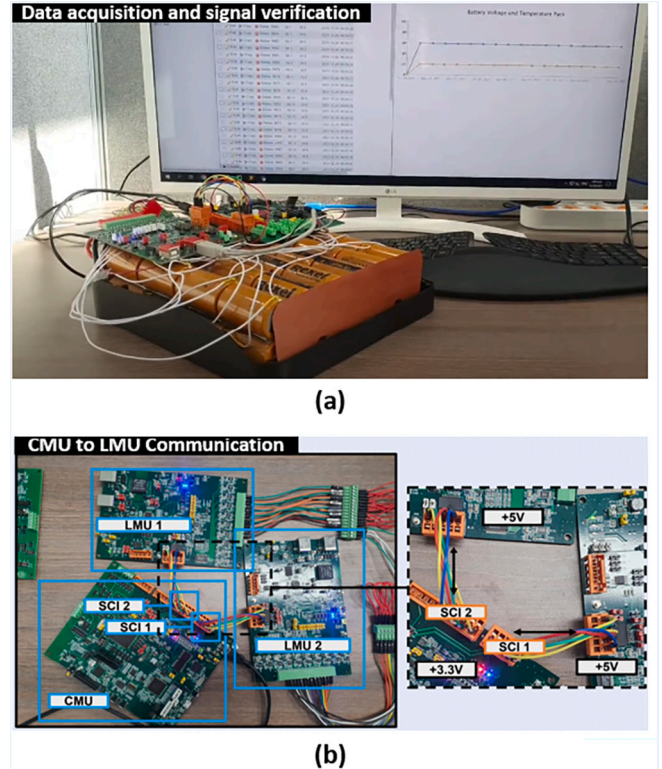


Fig. 7. Actual set-up (a) signal verification and acquisition (b) modular BMS connection.

conditions varied between nominal and full loads, simulating real-world scenarios to evaluate how the BMS adapts to different power demands. As a used-cased scenario, the nominal and full load pertain to the operating voltage of each battery cells which are 3.6 and 4.2 V respectively.

To demonstrate the effectiveness of the proposed modular BMS in fault scenarios, we present Fig. 8, which shows the implementation testing results. Fig. 8(a) illustrates four LMUs, each with four cells operating at a nominal voltage of 3.6 V, in normal conditions. Fig. 8(b) provides a magnified view of LMUs 2 and 3, showing that they continue to operate normally, unaffected by the faulty LMU 1. Fig. 8(c) presents

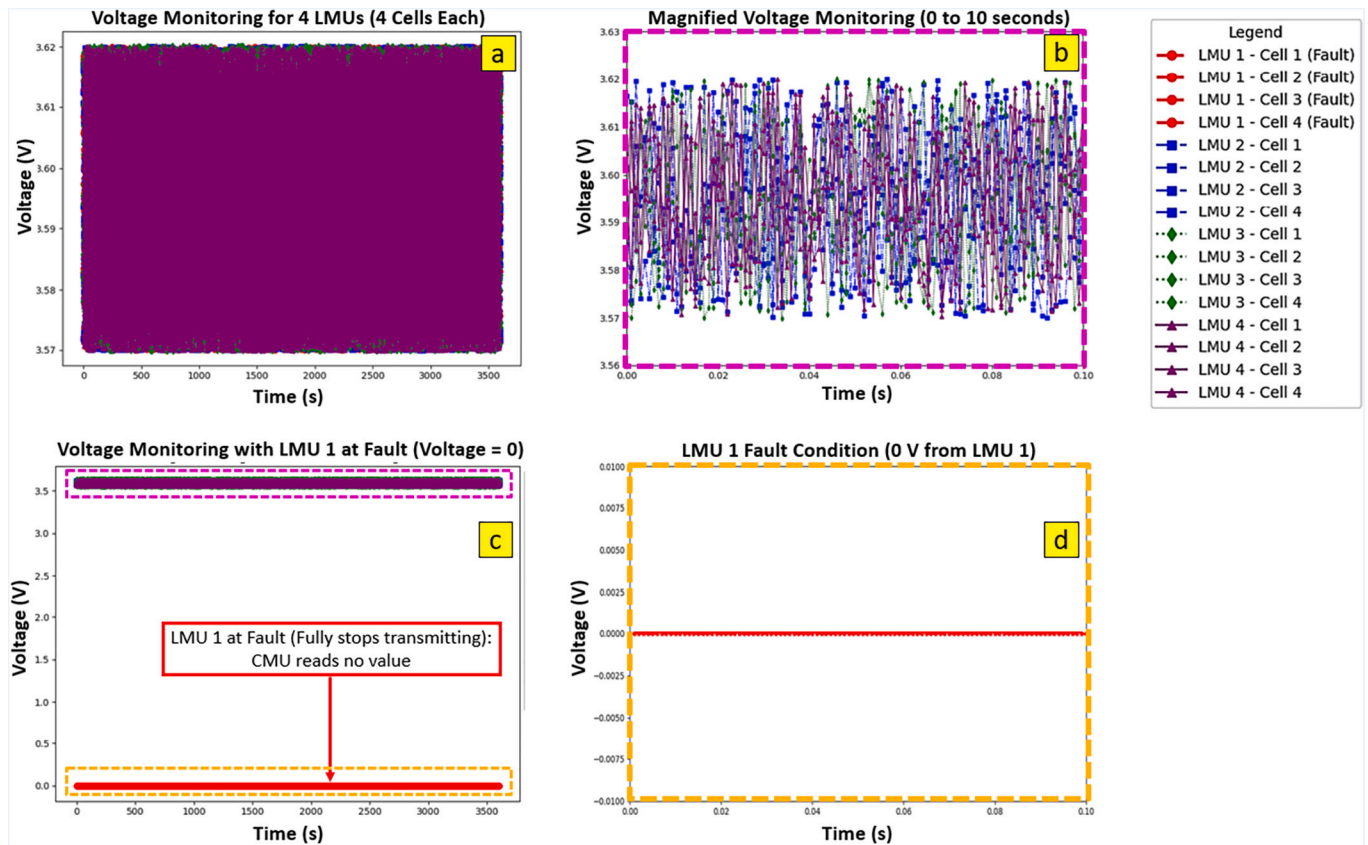


Fig. 8. Proposed modular BMS with hierarchical isolation (a) 4 LMUs having 4 cells at nominal operating condition (b) proposed BMS acquired voltage data when 1 LMU is at fault (LMU 1) (c) magnified version of LMUs 2–4 s' voltage monitoring when LMU 1 is at fault and (d) voltage reading from the LMU 1 as received by the CMU (LMU 1 at fault).

the case where LMU 1 is at fault, displaying 0 V for the entire operating point, confirming that the LMU has ceased functioning. Finally, Fig. 8(d) compares the voltage readings of LMU 1 (the faulty module) with LMUs 2 and 3, further demonstrating the isolation's fault tolerance. These waveforms validate the results presented in Tables I and II, proving that the proposed BMS effectively isolates faults while maintaining the stability of other modules. Compared with other BMS models [10–13] which yielded an average voltage difference of 0.185 V, the proposed BMS shows a significantly as highlighted in Table II, at just 0.002 V. Having a fair comparison is critical to prove the implementation viability and effectiveness of the proposed approach, reference [10–13] were chosen to highlight the stability of the modular BMS even when subjected to faulty condition. Remarkably, even when faults are present in the system, the proposed BMS achieves a voltage measurement accuracy that is 98.92 % higher than that of the existing models. It highlights that under faulty conditions, the proposed BMS not only maintains high accuracy but also offers superior deployment flexibility and enhanced safety.

4. Conclusion

In this paper, we introduce a hierarchical modular BMS with an advanced isolation architecture that enhances both flexibility and safety. The system integrates a CMU and multiple LMUs, providing adaptability for various battery pack configurations. The key innovation is the hierarchical isolation approach, which incorporates module-level, component-level, and ground-level isolation. This multi-tier isolation ensures faults are effectively contained, allowing the system to maintain stable operation and minimize the impact on unaffected modules.

The experimental results validate the robustness of the proposed BMS, demonstrating a mean absolute error of just 0.0015 V between the

actual and measured voltages—a 98.92 % improvement over conventional designs. Even under conditions where multiple LMUs experience faults, the system maintains high accuracy and voltage stability, underscoring the effectiveness of the hierarchical isolation in enhancing both fault tolerance and operational reliability.

CRedit authorship contribution statement

Henar Mike O. Canilang: Writing – original draft, Software. **Angela C. Caliwag:** Writing – original draft, Software, Conceptualization. **Wansu Lim:** Writing – review & editing, Writing – original draft, Supervision.

Funding

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (RS-2024-00349885).

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Wansu Lim reports financial support was provided by National Research Foundation of Korea. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] H.M.O. Canilang, et al., Design of modular BMS and real-time practical implementation for electric motorcycle application, *IEEE Trans. Circuits Syst. II Exp. Briefs* 69 (2) (Feb. 2022) 519–523.
- [2] T. Conway, et al., An isolated active balancing and monitoring system for lithium-ion battery stacks utilizing a single transformer per cell, *IEEE Trans. Power Electron.* 36 (4) (Apr. 2021) 3727–3734.
- [3] S. Maitreya, et al., Scalable and de-centralized battery management system for parallel operation of multiple battery packs, in: *Proc. IEMRE*, Mar. 2021, pp. 1–7 (Kolkata, India).
- [4] Y. Yang, K. Hu, et al., Digital battery management design for point-of-load applications with cell balancing, *IEEE Trans. Ind. Electron.* 67 (8) (Aug. 2020) 6365–6375.
- [5] G.R. Sylvestrin, et al., Hardware and software development of an open-source battery management system, *IEEE Lat. Am. Trans.* 19 (7) (July 2021) 1153–1163.
- [6] X. Zuo, et al., Flexible design for battery management system, in: *Proc. ICNISC*, 2016, pp. 168–172.
- [7] W. Guan, et al., A modular active balancing circuit for redox flow battery applied in energy storage system, *IEEE Access* 9 (2021) 127548–127558.
- [8] W. Kevin Siagian, et al., Design and implementation of battery management system for on-grid system, in: *Proc. ICCEREC*, May 2019, pp. 179–183 (Bandung, Indonesia).
- [9] M. Cherukuri, et al., Battery management system design for electric vehicle, in: *Proc. DISCOVER*, Aug. 2019, pp. 1–6 (Manipal, India).
- [10] Y. Xu, et al., Research and design of lithium battery management system for electric bicycle based on Internet of things technology, in: *Proc. CAC*, Hangzhou, China, Nov. 2019.
- [11] F. Zhu, et al., Battery management system for Li-ion battery, *J. Eng.* 2017 (13) (Jan. 2017) 1437–1440.
- [12] T. Yue, et al., High-precision voltage measurement IP core for battery management SoC of electric vehicles, in: *Proc. ICSICT*, Guilin, China, Oct. 2014.
- [13] B. Dimitrov, G. Collier, A. Cruden, Design and experimental verification of voltage measurement circuits based on linear optocouplers with galvanic isolation for battery management systems, *World Elec. Veh. J.* 10 (4) (Sep. 2019) 59.
- [14] R. Dutt, S. Sahu, A. Sarkar, A. Acharyya, Next-generation battery management system design methodology, in: *Proc. NEWCAS*, Edinburgh, United Kingdom, 2023, pp. 1–2.
- [15] G. Liang, G.G. Farivar, G.N.B. Yadav, E. Rodriguez, J. Pou, Battery fault tolerance of modular multilevel converter-based battery energy storage systems with redundant submodules, in: *Proc. Toronto*, Canada, 2021, pp. 1–6.